

Unit 6, Lesson 6: Similarity in Transformations



Benchmarks

MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



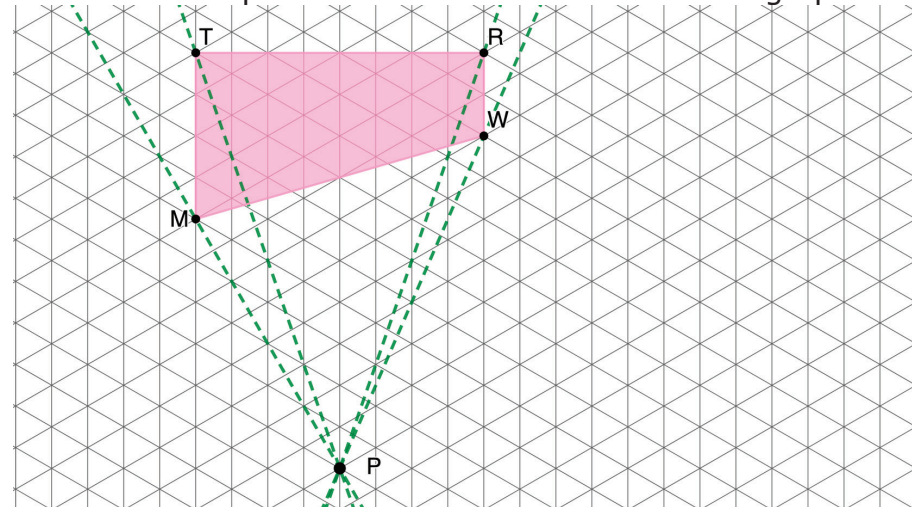
Learning Targets

- ☐ I can map one similar figure onto another and justify that the figures are similar.
- ☐ I can describe a sequence of transformations that will map the preimage onto the image.



6.6.1 Warm-Up: Dilation on an Isometric Graph

1. $TRWM$ and point P are drawn on an isometric graph.



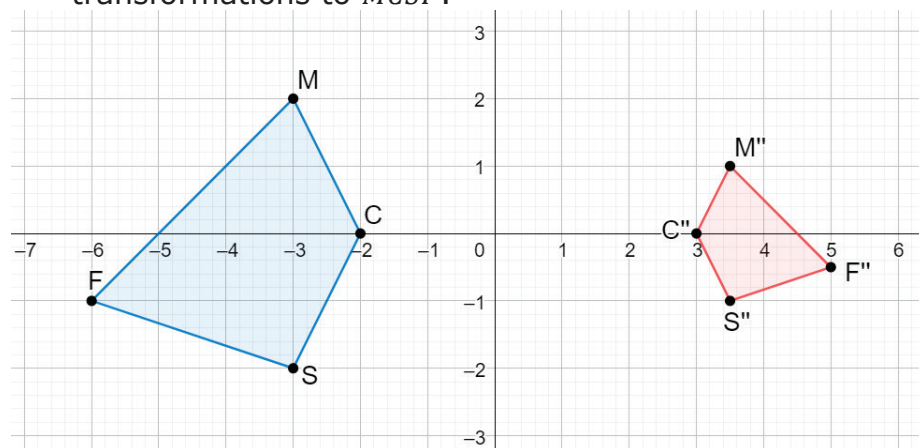
- a. Draw a dilation of $TRWM$ centered at point P using a scale factor of $\frac{1}{2}$.
- b. Explain what features on the graph were most useful in helping you determine where the dilation of $TRWM$ should be drawn.



6.6.2 Exploration: Similarity

With your partner, complete the following.

- $M''C''S''F''$ was created by applying a sequence of transformations to $MCSF$.



- With patty paper, trace $\angle FMC$ in $MCSF$.
 - Place the tracing over $\angle F''M''C''$ in $M''C''S''F''$.
 - What do you notice about $\angle FMC$ and $\angle F''M''C''$?
- Repeat the tracing for $\angle C$, $\angle S$, and $\angle F$. What do you notice about each angle and their corresponding angles?

- Complete the table.

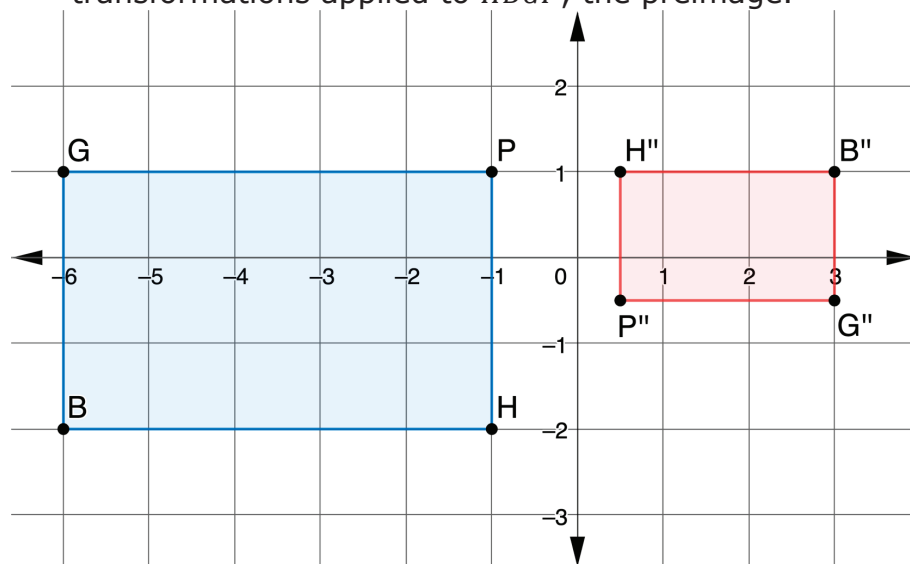
Angles	Comparison Between Angles
$\angle FMC$ and $\angle F''M''C''$	
$\angle MCS$ and $\angle M''C''S''$	
$\angle CSF$ and $\angle C''S''F''$	
$\angle SFM$ and $\angle S''F''M''$	

- Determine a sequence of transformations that can be used to map $MCSF$ to $M''C''S''F''$.
- Ask two classmates the transformations they determined. Record their transformations and write a summary of their reasoning. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating your summary was correct. Remind your classmates to include any details on the center of rotation or dilation and any scale factor.

Transformations	My Summary of Their Reason	Initials

- Work with your partner to verify your classmates' transformations are correct.

2. $H''B''G''P''$ was created using a sequence of transformations applied to $HBGP$, the preimage.



- a. Complete the table.

Preimage	Side Length	Image	Side Length	Ratio of Preimage to Image Lengths
$\overline{GP}$		$\overline{G''P''}$		
$\overline{PH}$		$\overline{P''H''}$		
$\overline{HB}$		$\overline{H''B''}$		
$\overline{BG}$		$\overline{B''G''}$		

- b. Describe the relationship between the ratios of the preimage to the image.

- c. Determine the sequence of transformations that was used to create $H''B''G''P''$ from $HBGP$.
- d. Discuss with your partner how to describe how $MCSF$ is related to $M''C''S''F''$ and how $HBGP$ is related to $H''B''G''P''$. Summarize your discussion.

3. What qualities of similar figures do these quadrilaterals have?



Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation or a sequence of transformations that includes a dilation.

4. Complete the statements using words from the bank.

Word Bank

congruent dilation proportional similar

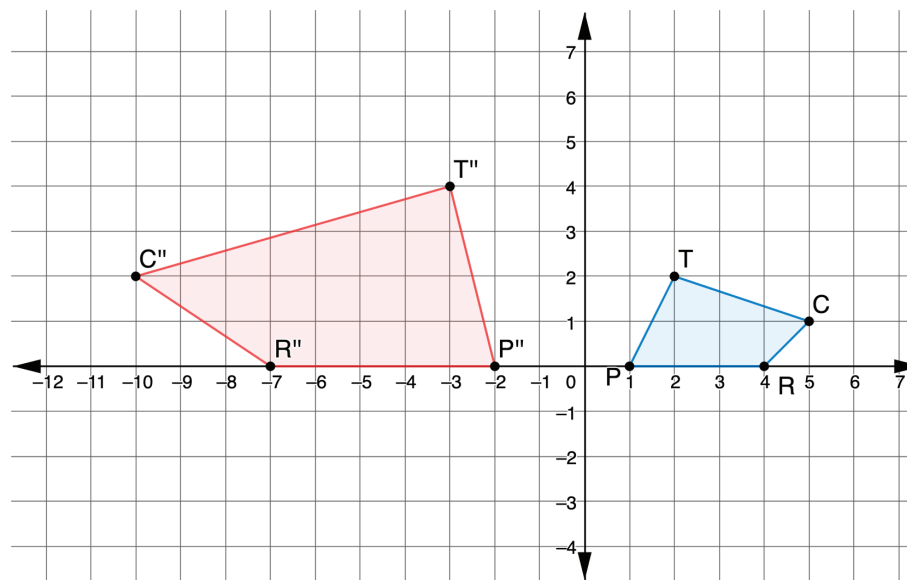
When a sequence of transformations includes a _____, the resulting image of the sequence of transformations is _____ to the preimage. All of the angles of the resulting image are _____ to the corresponding angles of the preimage. The corresponding sides of the image to the preimage are _____.



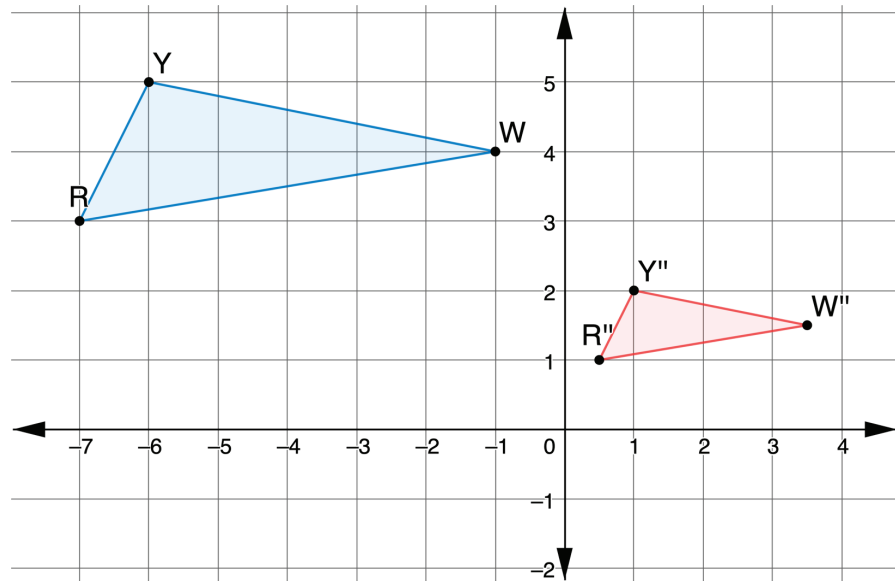
6.6.3 Guided Practice: Similarity

For each pair of figures, write the sequence of transformations that would show the figures are similar. If the figures are not similar, explain why.

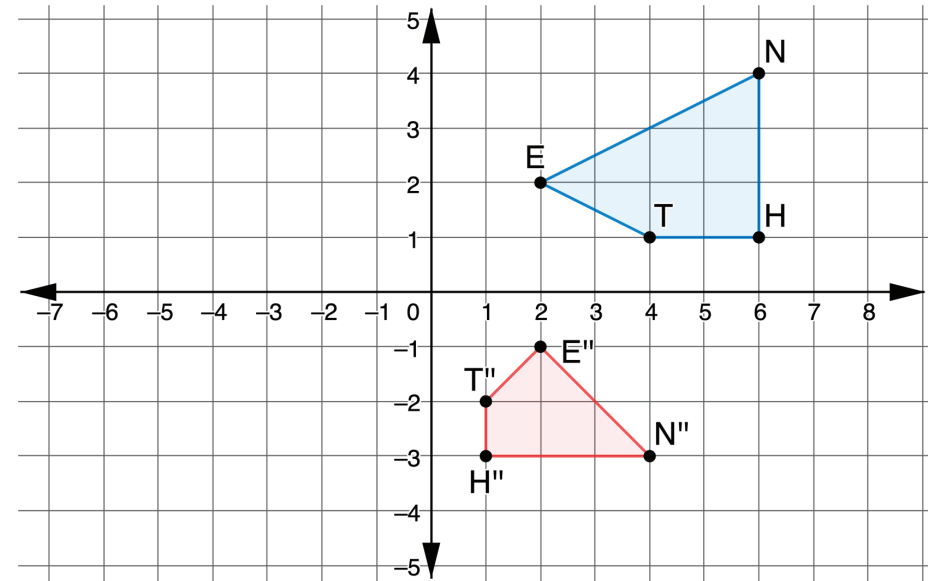
1.



2.



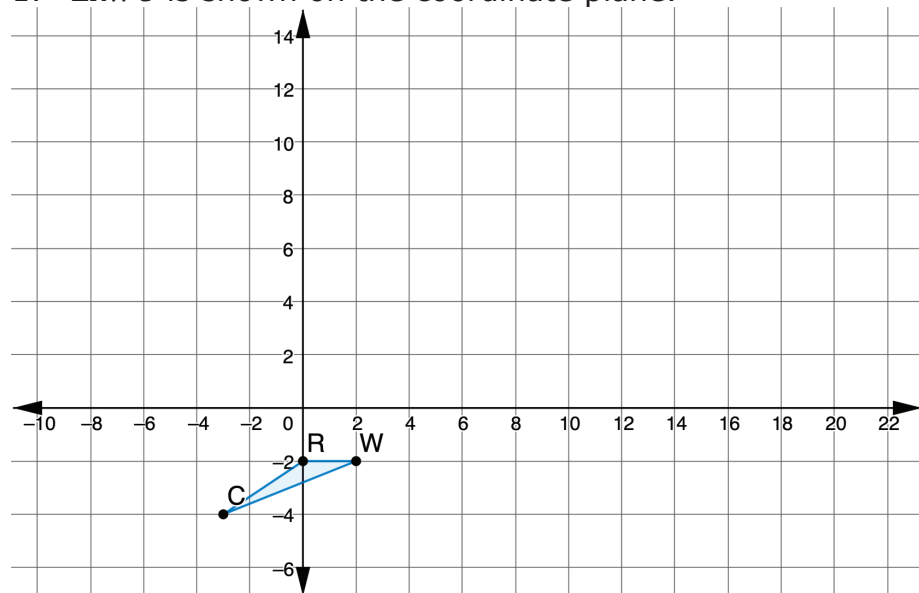
3.





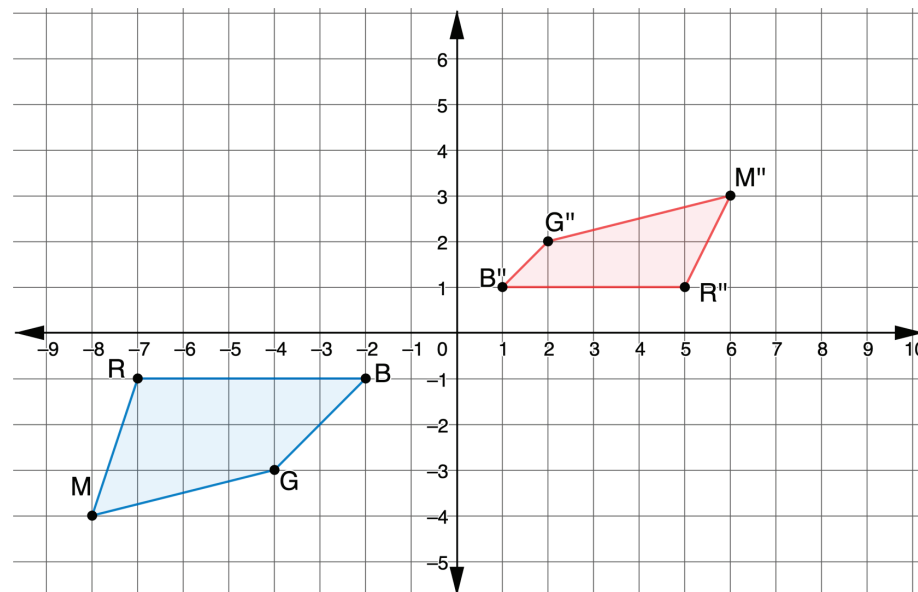
6.6.4 Your Turn!

1. $\triangle RWC$ is shown on the coordinate plane.



- Draw $\triangle R''W''C''$, which is a result of a translation up 4 units and left 1 unit, then dilated using a scale factor of 5 centered at C' .
- Explain how $\triangle RWC$ is similar to $\triangle R''W''C''$.

2. Carter and Isabella are asked to determine if the quadrilaterals are similar. Their responses are recorded in the table.



Carter's Response	Isabella's Response
Yes, the quadrilaterals are similar because all the corresponding angles are congruent.	No, the quadrilaterals are not similar because the corresponding sides are not proportional.

Is Carter's or Isabella's response correct? Explain your decision.



6.6.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand how to determine if a sequence of transformations results in similar figures.

Unit 6, Lesson H1: Proving All Circles are Similar



Benchmark

MA.912.GR.6.5 Apply transformations to prove that all circles are similar.



Learning Target

- ☐ I can justify that two circles are similar and conclude that any circle can be mapped to another circle using similarity transformations.